RELATIVE PSEUDO-COHERENCE OF A COMPLEX ON A SPACE OVER A RING TESTED AGAINST ALL PSEUDO-COHERENT PAIRINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12164-12172 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12174-12227, the complete author proof (52 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete, spaces more general). Closes L16132 (wave 14) as load-bearing input.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-characterize-pseudo-coherent' also exists in stacks-more-morphisms — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma derived-chow (B side, in-source); chapter lemma compute-Fourier-Mukai-for-derived-chow (L16215); Derived Categories of Spaces, lemmas cohomology-base-change and flat-proper-pseudo-coherent-direct-image-general; Derived Categories of Schemes, lemma pseudo-coherent-on-projective-space; More on Morphisms, lemma check-relative-pseudo-coherence-on-charts (L16132).

Source reference. Original lemma 09UI: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10834–10851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12115–12126. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSJ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12013–12030. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 09UU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12227–12227. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CSG: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12227–12227. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12227–12227. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CTN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12227–12227. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a ring. Let X be an algebraic space separated and of finite presentation over A. Let $K \in D_{QCoh}(\mathcal{O}_X)$. If $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every pseudo-coherent E in $D(\mathcal{O}_X)$, then K is pseudo-coherent relative to A (Definition 09UI).*

Proof. Assume $K \in D_{QCoh}(\mathcal{O}_X)$ and $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every pseudo-coherent E in $D(\mathcal{O}_X)$. Let $x \in |X|$. We will show that K is pseudo-coherent relative to A in an étale neighbourhood of x. This will prove the lemma by our definition of relative pseudo-coherence.

Choose n, Z, z, V, E as in Lemma 0CSJ. Denote $p : X \times \mathbf{P}^n \to X$ and $q : X \times \mathbf{P}^n \to \mathbf{P}^n_A$ the projections. Then for any $i \in \mathbf{Z}$ we have

$$\begin{aligned} & R\Gamma(\mathbf{P}^n_A, Rq_*(Lp^*K \otimes^{\mathbf{L}} E) \otimes^{\mathbf{L}} \mathcal{O}_{\mathbf{P}^n_A}(i)) \\ & = R\Gamma(X \times \mathbf{P}^n, Lp^*K \otimes^{\mathbf{L}} E \otimes^{\mathbf{L}} Lq^*\mathcal{O}_{\mathbf{P}^n_A}(i)) \\ & = R\Gamma(X, K \otimes^{\mathbf{L}} Rp_*(E \otimes^{\mathbf{L}} Lq^*\mathcal{O}_{\mathbf{P}^n_A}(i))) \end{aligned}$$

by Derived Categories of Spaces, Lemma 08IN. By Derived Categories of Spaces, Lemma 0CTN the complex $Rp_*(E \otimes^{\mathbf{L}} Lq^*\mathcal{O}_{\mathbf{P}^n_A}(i))$ is pseudo-coherent on X. Hence the assumption tells us the expression in the displayed formula is a pseudo-coherent object of $D(A)$. By Derived Categories of Schemes, Lemma 0CSG we conclude that $Rq_*(Lp^*K \otimes^{\mathbf{L}} E)$ is pseudo-coherent on $\mathbf{P}^n_A$. By Lemma 0CSK we have

$$Rq_*(Lp^*K \otimes^{\mathbf{L}} E)|_{X \times_A V} = R(W \to V)_* K|_W$$

Since $W \to V$ is a closed immersion into an open subscheme of $\mathbf{P}^n_A$ this means $K|_W$ is pseudo-coherent relative to A for example by More on Morphisms, Lemma 09UU. $\square$